10.5 Solution: (a) For EM wave with frequency ω, we have

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = i\omega \mathbf{B}.$$

It is well-known that the solution to the curl equation

$$\nabla \times \mathbf{A} = \mathbf{B}$$

can be written in a symmetric form as

$$\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r}.$$

Therefore,

$$\mathbf{E} = \frac{1}{2}\, i\omega\, \mathbf{B} \times \mathbf{r}$$

and the induced current is

$$\mathbf{J} = \sigma \mathbf{E} = \frac{1}{2}\, i\omega\sigma\, \mathbf{B} \times \mathbf{r}.$$

The magnetic dipole moment for this current is given by

$$\begin{aligned}\mathbf{m} &= \frac{1}{2}\int \mathbf{r} \times \mathbf{J}\, d^3x = \frac{i\omega\sigma}{4}\int \mathbf{r} \times (\mathbf{B} \times \mathbf{r})\, d^3x \\ &= \frac{i\omega\sigma}{4}\int \left[r^2\mathbf{B} - \mathbf{r}(\mathbf{r}\cdot\mathbf{B})\right] d^3x.\end{aligned}$$

The second term will have only non-zero component in the $\mathbf{B}$-direction. Thus,

$$\begin{aligned}\mathbf{m} &= \frac{i\omega\sigma}{4}\int B\, r^2 \sin^2\theta\, d^3x \\ &= \frac{i\omega\sigma}{4}\mathbf{B}\cdot 2\pi \int_0^R r^4\, dr \int_{-1}^{1} \sin^2\theta\, d(\cos\theta) \\ &= \frac{i\omega\sigma}{4}\mathbf{B}\cdot 2\pi \frac{R^5}{5}\frac{4}{3} = \frac{2\pi i\omega\sigma}{15}R^5\mathbf{B}.\end{aligned}$$

This is equivalent to the expression in the book as

$$\frac{Z_0}{\mu_0} = \frac{1}{\sqrt{\mu_0\epsilon_0}} = c,$$

and

$$\omega = ck.$$

(b) In the presence of both electric and magnetic dipole moments, the scattered electric field is

$$\mathbf{E}_{sc} = \frac{e^{ikr}}{r}\frac{k^2}{4\pi\epsilon_0}\left[4\pi\epsilon_0\frac{\epsilon_r - 1}{\epsilon_r + 2}E_0R^3(\hat{\mathbf{n}} \times \hat{\boldsymbol{\epsilon}}_0) \times \hat{\mathbf{n}} - \frac{i4\pi\sigma Z_0}{k\mu_0}(kR)^2\frac{R^3}{30}\frac{E_0}{c^2}\,\hat{\mathbf{n}} \times (\hat{\mathbf{n}}_0 \times \hat{\boldsymbol{\epsilon}}_0)\right].$$

where we have used the fact that

$$\mathbf{B} = \frac{1}{c}\hat{\mathbf{n}}_0 \times \mathbf{E}.$$

Therefore,

$$\mathbf{f}(\hat{\mathbf{k}}, \hat{\mathbf{k}}_0) = k^2\left[\frac{\epsilon_r - 1}{\epsilon_r + 2}R^3(\hat{\mathbf{n}} \times \hat{\boldsymbol{\epsilon}}_0) \times \hat{\mathbf{n}} + i\frac{Z_0\sigma}{k}(kR)^2\frac{R^3}{30}(\hat{\mathbf{n}}_0 \times \hat{\boldsymbol{\epsilon}}_0) \times \hat{\mathbf{n}}\right].$$

Compared to Prob. 10.4, we can see that

$$\begin{aligned}
\mathrm{Im}\Big[\hat{\boldsymbol{\epsilon}}_0^* \cdot \mathbf{f}(\hat{\mathbf{k}} = \hat{\mathbf{k}}_0)\Big] &= \mathrm{Im}\left[\frac{\epsilon_r - 1}{\epsilon_r + 2} k^2 R^3 + i Z_0 \sigma k (kR)^2 \frac{R^3}{30}\right] |\hat{\boldsymbol{\epsilon}}_0 \times \hat{\mathbf{n}}_0|^2 \\
&= k^2 R^3 \frac{3 Z_0 \sigma / k}{(\epsilon_r + 2)^2 + (Z_0 \sigma / k)^2} + Z_0 \sigma k (kR)^2 \frac{R^3}{30}.
\end{aligned}$$

The total cross section is given by the optical theorem,

$$\begin{aligned}
\sigma_t &= \frac{4\pi}{k} \mathrm{Im}\Big[\hat{\boldsymbol{\epsilon}}_0^* \cdot \mathbf{f}(\hat{\mathbf{k}} = \hat{\mathbf{k}}_0)\Big] = 4\pi \left[R^3 \frac{3 Z_0 \sigma}{(\epsilon_r + 2)^2 + (Z_0 \sigma / k)^2} + Z_0 \sigma (kR)^2 \frac{R^3}{30}\right] \\
&= 12\pi R^2 (R Z_0 \sigma) \left[\frac{1}{(\epsilon_r + 2)^2 + (Z_0 \sigma / k)^2} + \frac{(kR)^2}{90}\right].
\end{aligned}$$